CyberShake physics-based seismic hazard maps using reduced-order models: results from Southern San Andreas earthquake scenarios

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Key Points

- Physics-based probabilistic seismic hazard analysis is computationally expensive for many faults.
- Our reduced-order model efficiently produces high-resolution hazard corrections for 480,000 sites.
- CyberShake calculations may incorporate forward simulations to improve ground motion resolution.

Keywords

computational seismology, reduced-order model, CyberShake, seismic hazard, machine learning

Abstract

Recent advances in Probabilistic Seismic Hazard Analysis (PSHA) leverage physics-based ground motion simulations to estimate seismic hazard, such as the CyberShake project. However, computational costs quickly escalate when performing PSHA for numerous faults or sites and can become prohibitively expensive. To reduce computational demands, hazard maps commonly interpolate ground shaking intensities from simulations conducted at fewer locations, but the accuracy of these interpolations remains poorly quantified. To address these questions, we generate a set of earthquake point source simulations distributed along the non-planar UCERF3 fault geometry for the Southern San Andreas Fault (SSAF) extending from Bombay Beach to Lake Hughes. Using SeisSol, we simulate three minutes of viscoelastic seismic wave propagation for these sources and use a parallel HDF5 file format to store horizontal-component velocity seismogram data for 480,000 sites. From these data, we derive frequency-dependent site terms for our high-resolution sites in Southern California, as well as the CyberShake Study 22.12 sites. Using minimum curvature interpolation with tension as used in CyberShake, we compare the interpolated site terms against our high resolution site terms. We identify local discrepancies for our set of earthquake sources with peak ground velocities differing by up to a factor of approximately three. Furthermore, we identify areas where unexpectedly high or low ground motions could be missed when using the interpolated dataset to create a hazard map. To generalize our results across broader kinematic rupture models for the SSAF scenarios, we utilize an accurate reduced-order modeling technique, a scientific machine learning approach based on interpolated proper orthogonal decomposition, to rapidly compute Green's functions. The approach may be used within CyberShake to reduce the time-to-solution by an estimated factor of 336. Our analysis of physics-based site terms from ensembles of SSAF rupture scenarios more accurately quantifies seismic hazard and guides future developments combining high-performance computing and machine learning reduced-order modeling techniques for PSHA.

Introduction

To prepare for the damaging effects of earthquakes, probabilistic seismic hazard analysis (PSHA, refer to the Appendix for the full list of acronyms) quantifies the probabilities of exceeding certain levels of ground motion. PSHA is used worldwide to create seismic hazard models (Petersen et al., 2024; Gerstenberger et al., 2024). These hazard models inform policy on seismic design spectra for the construction of buildings and infrastructure, help decide which structures should be prioritized for retrofitting, and determine insurance rates. PSHA considers a set of possible earthquakes, their recurrence intervals, and the ground motions produced by each possible earthquake. An emerging direction in PSHA is to use physics-based simulations of seismic wave propagation to determine ground motion, as in the CyberShake project (Graves et al., 2011). Due to the limitations of available ground motion data and the

ergodic assumption when using empirical ground motion models, which assumes that spatial variability across sites can be represented by temporal variability observed at a single site (Stafford, 2014), we instead quantify hazard using physics-based earthquake ground motion simulations. It is also possible to consider a physics-based perspective from the generation of physics-based rupture models, such as by using dynamic rupture and seismic cycling models (Shaw et al., 2025; Milner et al., 2021; Shaw et al., 2018), though this is a separate topic that we do not consider in this paper. Evidence suggests that most sites display 3D response behavior, and not 1D response, implying that empirical corrections may not be sufficient to express the site response correction using simple functional forms (e.g., Thompson et al., 2012; Rodgers et al., 2020). Significant effort devoted towards developing high-resolution seismic velocity and attenuation models of the near-surface and crust motivates the use of these models in physics-based PSHA.

Since a CyberShake study typically considers far more earthquakes than recording sites, usually by a factor of around 1000, a reciprocity-based approach is used so that the number of simulations scales with the number of sites rather than the number of events (Zhao et al., 2006). This approach computes strain Green's tensors (SGTs) by placing an impulse at the site of interest and performing three anelastic finite-difference wave propagation simulations, one for each component. SGTs are saved for each point in the fault geometry representation. The next step generates seismograms for each rupture through a convolution of the corresponding SGTs and the slip time history at each point along the surface of the rupture. These seismograms can be processed to obtain derived data products such as spectral acceleration, specific durations, site-specific hazard curves, and regional hazard maps (Graves et al., 2011; Jordan and Callaghan, 2018; Callaghan et al., 2024). The state-of-the-art approach computes the site terms at a number of pre-selected locations by computing the strain Green's tensors for a set of candidate faults or source locations. The site terms are then interpolated using continuous curvature splines with tension to eliminate large oscillations (Smith and Wessel, 1990) and applied as a correction to the map based on the GMMs.

Previous work has examined approaches to speed up the CyberShake workflow. For example, Azar and Dabaghi (2021) looked to speed up CyberShake calculations by using a Monte-Carlo simulation approach that looks at finite length earthquake catalogs. This gives substantial speed-up in the post-processing step which scales with the product of the number of time steps and the number of rupture surface points involved in the convolution process. However, the proposed approach has little impact on the computational cost of the SGT calculation step. Other efforts optimizing high-performance computing (HPC) codes and the use of GPUs for the wave propagation simulations also help to speed up the simulations (Cui et al., 2013). However, as the computational cost of hazard analysis continues to increase with both the maximum frequency and the number of sites, we seek a solution that has better scaling with the number of sites.

To reduce this computational cost, we explore developments in scientific machine learning and reduced-order modeling to improve the computational efficiency of physics-based PSHA. Here, we use physics-based in the sense that the ground motions are estimated from seismic wave propagation simulations, and not from GMMs. Specifically,

we use reduced-order models (ROMs) of complex 3D seismic wave propagation (Rekoske et al., 2025) to significantly speed up the simulations. The full-order model (FOM), described in the Numerical method section, refers to solving the discretized, partial differential equation governing the anelastic seismic wave propagation. For the ROM, we use a modified version of the Rekoske et al. (2025) algorithm to calculate Fourier amplitude spectra (FAS) for a set of 163 logarithmically spaced frequencies from 0.1 to 1.0 Hz. Using the method, we dramatically increase the number of sites where we compute seismograms, up to 480,000 compared to the 335 used in both the CyberShake 21.12 and 22.12 studies in Southern California. This facilitates more detailed study on the seismic wavefield and its effects on structures without incurring a high computational cost.

The overview of this paper is as follows: The Methods section describes the methods to determine the locations and properties of the point sources we use for simulations. The methods section also describes the forward simulation approach using SeisSol, the kinematic rupture model generator, and the reduced-order modeling using computed Green's functions to obtain Fourier amplitude spectra for all the rupture models. In the Results section, we show the results comparing our high-resolution maps of hazard correction terms for both point source and kinematic rupture models against a lower-resolution set of sites following the CyberShake interpolation approach using minimum curvature interpolation with tension. The paper ends with a discussion in the Discussion that lays out ways these results could inform or improve future hazard studies, and how the results could be expanded to consider statewide California earthquake rupture forecasts and additional intensity metrics.

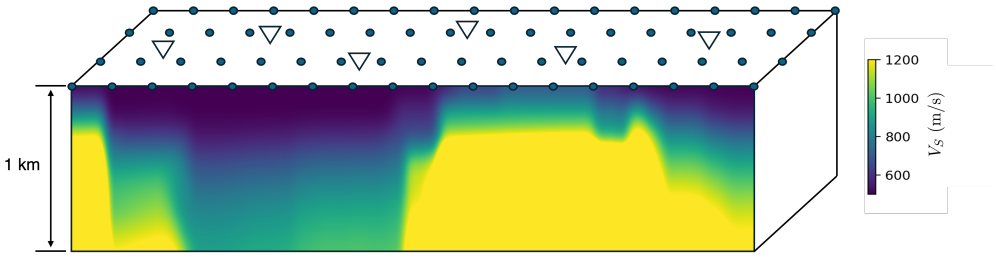


Figure 1: Conceptual figure to illustrate the motivation to improve the spatial resolution in physics-based seismic hazard assessment. Assuming a crustal velocity model, as shown in the cross-section for a selected vertical slice through the Southern California CVM-S4.M26.01, heterogeneities in the subsurface V_S structure result in many different velocity profiles. Adequate resolution of receivers at the surface is required to capture the wavefield effects and resulting hazard. Lower resolution maps, such as using the triangles as sites, are computationally cheaper but not as accurate as using higher resolution sites shown by the blue circles.

Alt-text: Cross-section showing subsurface shear-wave velocity structure. Surface receivers shown as triangles (low resolution) and blue circles (high resolution). Velocity heterogeneities visible as varying colors with depth.

Methods

Source selection along the Southern San Andreas Fault using a Halton sequence

In Southern California, the Southern San Andreas Fault (SSAF) last produced a large earthquake in 1857, and is estimated to have a mean 53% chance (min. 17%, max. 93%) of producing an $M_W > 6.7$ earthquake in the next 30 years (Field et al., 2015). Quantifying the range of expected ground motions is thus important for PSHA. The extent of SSAF ruptures that has been considered in recent hazard calculation scenarios has varied. For example, the ShakeOut scenario considers a rupture starting at Bombay Beach in the south, and ending at Lake Hughes in the north (Graves et al., 2011). Here, we use a Halton sequence (Halton, 1960) of $n_h = 100$ point sources placed along this part of the SSAF, using the UCERF3 fault geometry (Field et al., 2014) following the same extent at the 2008 ShakeOut source model (Figure 2). We use a Halton sequence to efficiently sample across the length and width of the fault, and we find that using $n_h = 100$ provides us with sufficient sampling density to construct the ROM for seismic waveform modeling (see more details in the Numerical method section). Each point $\boldsymbol{p}_i$ is expressed in 2D as (Δx_i, Δz_i) where Δx_i is the distance from a reference point at the northwest-most part of the fault at Lake Hughes, and Δz_i is the hypocentral depth. The source depths range from 0.15 km to 15 km, and the total length of the fault that we consider is approximately 340 km. We choose 0.15 km as the minimum depth as this equals the shallowest depths used in the CyberShake calculations of the strain Green's tensors. Although some faults in the UCERF3 source model are defined to reach the surface, the locations for the Green's functions calculations are the center of each subfault, meaning that the center is $\frac{1}{2}$ times the corresponding subfault dimension.

We then transform these parameters $\boldsymbol{p}_i$ to the actual locations in 3D space using the UCERF3 fault geometry. This source model is not the same as the CyberShake 22.12 study, which instead considers an ensemble of rupture variations using fault surfaces from the UCERF2 model. In the San Gorgonio Pass region, the SSAF consists of complex faulting along multiple strands (Beyer et al., 2018). We do not include any sources on the northern strands, such as the Mill Creek North Branch, which would result in identical $\boldsymbol{p}_i$ values when their 3D location along is mapped to the Δx_i and Δz_i parameters. The consideration of rupture possibilities on multiple branches is also important for PSHA but remains outside the scope of the current study. Our moment tensor at each point uses a strike equal to the average of 122 degrees, a dip of 90 degrees, and a rake of 0 degrees. This method does not account for any non-strike-slip motion, and also does not account for any change in the fault dip along the strike. We point out here that these are simplifications to reduce the number of simulations that must be performed, though future work could use additional representative moment tensors (e.g., Kikuchi and Kanamori, 1991). We further discuss the additional computational costs that we would incur to address these limitations in the Discussion section.

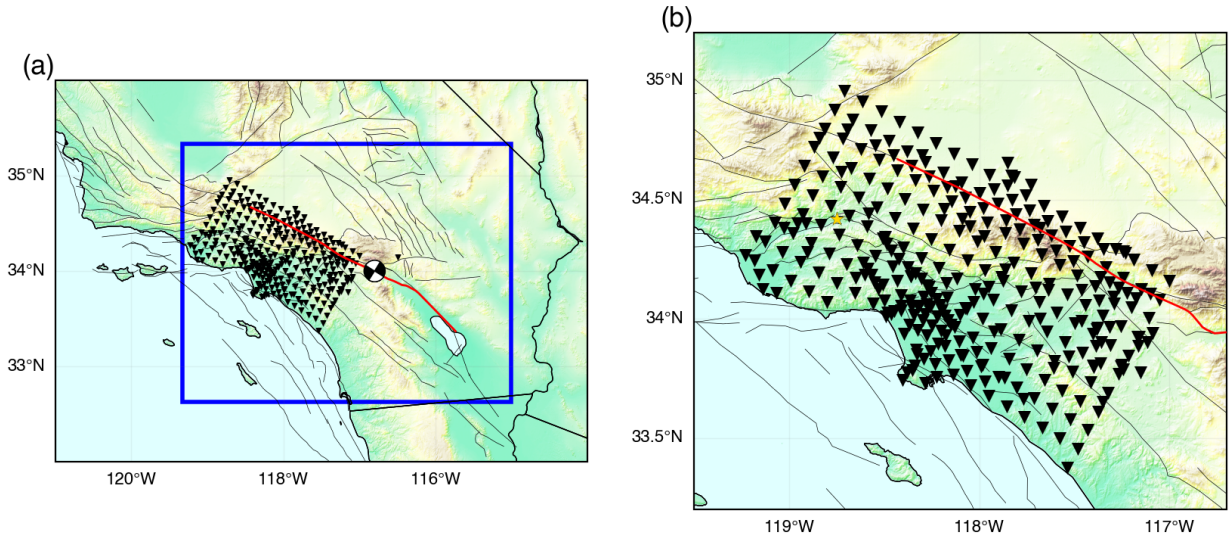


Figure 2: Maps of the study area in Southern California. (a) shows the UCERF3 fault traces indicated by gray lines, and the portion of the SSAF that we use is shown in red. The blue rectangle indicates the boundary of the area where we compute high resolution ground motion site terms for the SSAF ruptures. The CyberShake Study 22.12 sites in Southern California are indicated by the black triangles. (b) shows a zoomed in view of the CyberShake Study 22.12 sites. The gold star indicates a location in the narrow Fillmore Basin where hazard is challenging to model accurately.

Alt-text: Two-panel map of Southern California. Panel (a) shows UCERF3 fault traces as gray lines with red Southern San Andreas Fault segment, blue study area rectangle, and black triangles marking CyberShake sites. Panel (b) zooms into CyberShake sites with gold star marking Fillmore Basin location.

Full-order model simulations using SeisSol

At each of the source locations, we perform seismic waveform modeling of point sources using the viscoelastic implementation in SeisSol (Uphoff and Bader, 2016), a high performance wave propagation and dynamic rupture code which uses the arbitrary high-order accurate derivative discontinuous Galerkin (ADER-DG) method (Käser and Dumbser, 2006; Käser et al., 2007; Dumbser and Käser, 2006; Heinecke et al., 2014; Rettenberger et al., 2016; Uphoff and Bader, 2016; Uphoff et al., 2017). For more details regarding the viscoleastic implementation in SeisSol please refer to details in Uphoff and Bader (2016) and our model setup which follows Rekoske et al. (2025). For the simulations, we use a seismic velocity model following the CyberShake 22.12 approach which takes the minimum V_S between the CVM-S4.M26.01 model and the Ely et al. (2010) V_{S30} tapering method down to a depth of 700 m. V_{S30} is the time-averaged shear wave velocity in the uppermost 30 m. To obtain the seismic velocities, we use the UCVM software (Small et al., 2017) and the V_{S30} map by Thompson et al. (2014) using the v2.0 update (Thompson, 2022). We set a floor for the minimum S-wave speed at 500 m/s as used by CyberShake and preserve the original V_P/V_S ratio. We use 3D spatially variable anelastic attenuation factors of $Q_S = 0.05V_S$ and $Q_P = 0.1V_S$ (for V_S measured in m/s) to describe the attenuation of seismic waves (Olsen et al., 2003). We then use velocity-aware meshing (Breuer and Heinecke, 2022) to create a tetrahedral mesh resolving frequencies up to 1.0 Hz for the whole domain area shown in Figure 2 with a maximum depth of 20 km. The computational mesh extends beyond the area of interest with gradual element coarsening to avoid artificial reflections from the boundaries. The size of the mesh is 8,016,918 elements. We run the simulations on Frontera using single-precision floating point numbers and using a degree six polynomial in space and time. The computational cost per simulation is 3,920 CPU hours (CPUh), or 70 node-hours.

From the SeisSol simulations, we save the wavefields of the velocity seismograms with a spatial sampling of 500 meters in the area of interest (Figure 2), and a sampling rate of 5.0 Hz. With 500 m spatial sampling, we have a total of $n_r = 480,000$ receiver locations. Using the velocity seismograms, we then compute the time derivative to obtain acceleration and then take the Fast Fourier Transform of the acceleration timeseries using the algorithms available in numpy (Harris et al., 2020). To obtain the orientation independent effective amplitude spectrum (EAS) defined by Goulet et al. (2018), we take the quadratic mean of the two horizontal components,

$$\mathrm{EAS}(f) = \sqrt{\frac{1}{2}\left[\mathrm{FAS}_1(f)^2 + \mathrm{FAS}_2(f)^2\right]}, \tag{1}$$

where FAS_1 and FAS_2 indicate the Fourier amplitude spectra for two orthogonal components processed following the Pacific Earthquake Engineering Research Center's processing procedure (Kishida et al., 2016).

Finite fault simulation approach

For PSHA with large earthquakes, it is important to consider the finite rupture effects, including directivity. To generate appropriate rupture models for our analysis, we use version 5.5.2 of the Graves and Pitarka (2016) kinematic rupture generator, which provides a way to generate ruptures with stochastic slip and rise time variations. The ShakeOut scenario uses a hypocenter location at the southern end at Bombay Beach, as this is thought to be a likely nucleation point for SSAF earthquakes (Stuart, 1986; Hudnut et al., 1989). However, the hypocenter location for these future earthquakes is unknown, and it is important for seismic hazard analysis to consider how different nucleation points and directivity effects could affect the scenario outcomes and other cascading earthquake effects. Thus, we generate 100 unique kinematic rupture models for a set of uniformly distributed hypocenter locations. We do not allow perturbations of the subfault locations, so all the slip occurs directly on the UCERF3 fault geometry. The complete list of input parameters to the rupture generator appears in Table 1 in the Supplemental Material. For one of the generated ruptures, shown in Figure 3, we then simulate the seismograms and compute the EAS. We quantify the ROM and FOM agreement for this single kinematic rupture model shown in Figure 3, but we do not perform this analysis for all the rupture models as this would require performing additional expensive simulations for each rupture model.

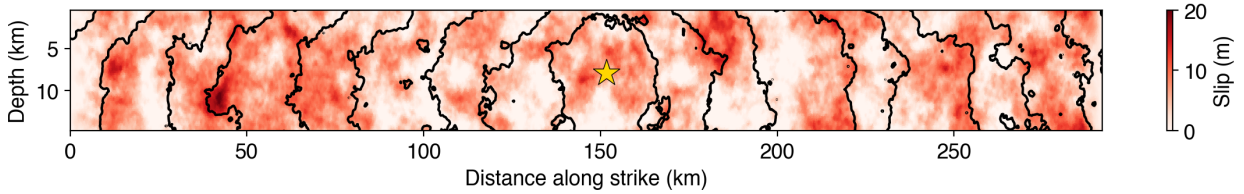


Figure 3: A single kinematic rupture model for the SSAF generated using v5.5.2 of the Graves and Pitarka (2016) kinematic rupture generator. The hypocenter location is indicated by a gold star. The black contours indicate the initiation time of slip in increments of 5 s. 99 other stochastic kinematic rupture models for the same fault geometry are also generated with varying hypocentral locations.

Alt-text: Map view of Southern San Andreas Fault kinematic rupture model showing slip distribution as colored contours. Gold star marks hypocenter location. Black contour lines indicate rupture initiation time at 5-second intervals propagating along fault.

Parallel HDF5 data format for high-resolution seismogram storage

To efficiently store the large volume of data generated for this study, we use a parallel HDF5 data format. SeisSol supports writing data using individual ASCII files for receiver data, using one file per receiver location, but this option

is not preferred when the number of receivers becomes very large due to it becoming too slow for the file system. We instead implemented a parallel HDF5 writer using the HDF5 C++ API (Folk et al., 2011). Our approach was designed to be robust enough such that it can handle node failures. This HDF5 format for the output seismograms in SeisSol is planned to be released in a future SeisSol version. For information on accessing the data generated for this study, refer to the Data Availability Statement.

Reduced-order modeling using interpolated proper orthogonal decomposition

Using the east-component and north-component Fourier amplitude spectra, we compute the Green's functions by dividing the Fourier transform of the accleration seismograms, FAS_j, by the Brune-like moment rate spectra used for the point source simulations. We only keep the spectra for frequencies from 0.1 to 1.0 Hz, giving us a total of $n_f = 163$ frequencies to consider. The Green's functions are then gathered into two matrices $\boldsymbol{G}_j$, one for each horizontal component, where each row of $\boldsymbol{G}_j$ contains the data for a single receiver and frequency, and the column indicates the source location point on the SSAF. Thus, the size of each $\boldsymbol{G}_j$ is $(n_r n_f \times n_h)$. Each element of $\boldsymbol{G}_j$ is represented by a complex double precision floating point number.

We use the method from Rekoske et al. (2023) and Rekoske et al. (2025) to construct a reduced-order model using the interpolated proper orthogonal decomposition (POD) approach (e.g., Bui-Thanh et al., 2003; Druault et al., 2005). In this method, the first step is to compute the singular value decomposition (SVD) of each $\boldsymbol{G}_j$ matrix. In contrast to Rekoske et al. (2025) and Rekoske et al. (2023), which uses velocity seismograms in the $\boldsymbol{G}_j$ matrices, we instead choose here to insert the Green's functions into the reduced-order modeling framework. This helps to save computing Fourier transforms in the online stage, helping to make the ROM faster for predicting the spectral amplitudes. Thus, we compute the SVD of the data matrix,

$$\boldsymbol{G}_j = \boldsymbol{U}_j \boldsymbol{\Sigma}_j \boldsymbol{V}_j^T, \tag{2}$$

where $\boldsymbol{G}_j$ contains Green's functions for the two horizontal components ($j = 1, 2$). We compute the SVD in Eq. (2) using the eigendecomposition of $\boldsymbol{G}_j^T \boldsymbol{G}_j$ and then construct the radial basis functions to interpolate the POD coefficients. With this approach, the predicted Fourier amplitude spectra for finite rupture models are calculated with the ROM using

$$\tilde{\boldsymbol{q}}_j = \sum_{k=1}^{n_s} \dot{M}_k \tilde{\boldsymbol{G}}_j(\boldsymbol{p}_k) \tag{3}$$

where n_s is the number of subfaults in the kinematic rupture model, and $\dot{M}_k$ is the moment rate spectrum for the k-th subfault from 0.1 to 1.0 Hz. As we only consider a single strike-slip focal mechanism, which is aligned with the average fault strike as shown in Figure 2, we do not sum over all six elementary moment tensors which are necessary for general faulting styles (Kikuchi and Kanamori, 1991). Note that the predicted spectra will be complex-valued

containing both amplitude and phase, and we again take the absolute value and multiply by the sampling interval in time. The approximate Green's functions, $\tilde{\boldsymbol{G}}_j$, for the two horizontal components are obtained using

$$\tilde{\boldsymbol{G}}_j(\boldsymbol{p}^*) = \sum_{k=1}^{n_h} \tilde{\boldsymbol{A}}_{jk}(\boldsymbol{p}^*)\boldsymbol{u}_{jk}, \tag{4}$$

where $\tilde{\boldsymbol{A}}_{jk}$ are the interpolated POD coefficients using Eq. (11) from Rekoske et al. (2025) and $\boldsymbol{u}_{jk}$ are the left singular vectors from the $\boldsymbol{U}_j$ matrix for the k-th mode.

Results

High-resolution ground motion maps

To examine the quality of the interpolations of physics-based site terms, we first examine the high-resolution maps of EAS from the simulations for a point source and a single kinematic rupture model. The point source results are shown in Figure 4 and the kinematic rupture model results are shown in Figure 5. In each of these figures, we also show the median EAS predictions from the BA18 (Bayless and Abrahamson, 2019) ground motion model (GMM) which was developed for shallow crustal earthquakes in California. The BA18 GMM uses the V_{S30} and $Z_{1.0}$ site parameters to model the site response, and we obtain these parameters using the same velocity model that we use for the physics-based simulations. The radiation pattern is especially noticeable at this frequency of 0.5 Hz in the simulation results, though the GMM does not model this pattern as it uses data from a combination of focal mechanism orientations. We compute the residuals between simulation and GMM EAS predictions using the natural logarithm of the simulated EAS divided by the GMM. That is,

$$r(f) = \ln \mathrm{EAS}_{\mathrm{FOM}}(f) - \ln \mathrm{EAS}_{\mathrm{GMM}}(f) \tag{5}$$

where $r(f)$ is the EAS residual, $\mathrm{EAS}_{\mathrm{FOM}}(f)$ is the simulated EAS calculated in Eq. (1) and $\mathrm{EAS}_{\mathrm{GMM}}(f)$ is the GMM EAS.

The residuals computed in Eq. (5) represent the combined effects of event and site contributions. To isolate the site-specific effects, we remove the mean residual for each event across all sites. As a result, this approach then gives the centered spatially-varying residuals that represent the relative site-specific ground motion amplification or de-amplification. The centered residual term, $s(f)$, is computed as

$$s(f) = r(f) - \overline{r(f)}_{\mathrm{event}}, \tag{6}$$

where $\overline{r(f)}_{\text{event}}$ is the mean of the residuals for a given event computed across all sites. Thus the site terms are defined relative to an event-averaged baseline which could then be incorporated appropriately into the hazard maps.

We then pull a subset of receivers from our high-resolution grid of receivers, using only the receivers located the nearest to each CyberShake 22.12 station. We then spatially interpolate the centered residuals $s(f)$ at these receivers using GMT's surface algorithm (Wessel et al., 2019; Tian et al., 2025) which uses a minimum curvature interpolation with tension (Smith and Wessel, 1990) by solving

$$(1-T)\nabla^2(\nabla^2 z) - T\nabla^2 z = s(f), \tag{7}$$

where T is an adjustable tension parameter between 0 and 1, ∇^2 is the Laplacian operator, $s(f)$ is the data (in this case the centered residuals) and z is the coefficient associated with the minimum curvature interpolant. Using $T > 0$ can help suppress undesired oscillations or false local minima and maxima (Wessel et al., 2019). We solve Eq. (7) using a grid spacing of 0.02 degrees, an interior tension equal to 0, an exterior tension equal to 0.1, the default convergence tolerance, and a search radius of 20 arc minutes. These values were determined in CyberShake studies through a small parameter search and the resulting maps were not found to be sensitive to modest changes in these parameters. In CyberShake seismic hazard maps, the interpolation of the site terms is used as a modification to the hazard maps based on the ground motion models. For example, CyberShake Study 22.12 used their simulations and computed the residuals from ground motion models. The residuals are then interpolated and added onto the GMM estimates. Similarly, we use our high-resolution wavefields and select only the sites that correspond to the sites that CyberShake Study 22.12 uses.

Interpolation errors for point source and the finite rupture models

By interpolating the residuals at the CyberShake stations, we obtain the smooth spatial fields plotted in Figure 4d and Figure 5d. These show examples of the spatial effects that would be available upon using the CyberShake station data. However, the smooth fields do not capture the more detailed parts of the wavefield. Taking the difference between the high-resolution simulated residuals and the interpolated residuals, we quantify the error that results from spatially interpolating the residuals using data from a limited number of stations (Figures 4e and 5e). By definition, the interpolation error is equal to zero at points in the map where a CyberShake site is located. However, at points in between the stations, the difference in residuals can be up to ≈ 3 natural log units for both the point source and the kinematic rupture model results at 0.5 Hz. The largest interpolation errors spatially coincide with sharp lateral gradients in basin depths and topography (see maps in Figure S1), indicating that the minimum-curvature interpolation smooths over physically meaningful wavefield effects visible in the high-resolution simulations. Despite the additional complexity of rupture directivity and slip heterogeneity, the finite-fault results show that the interpolation error magnitudes

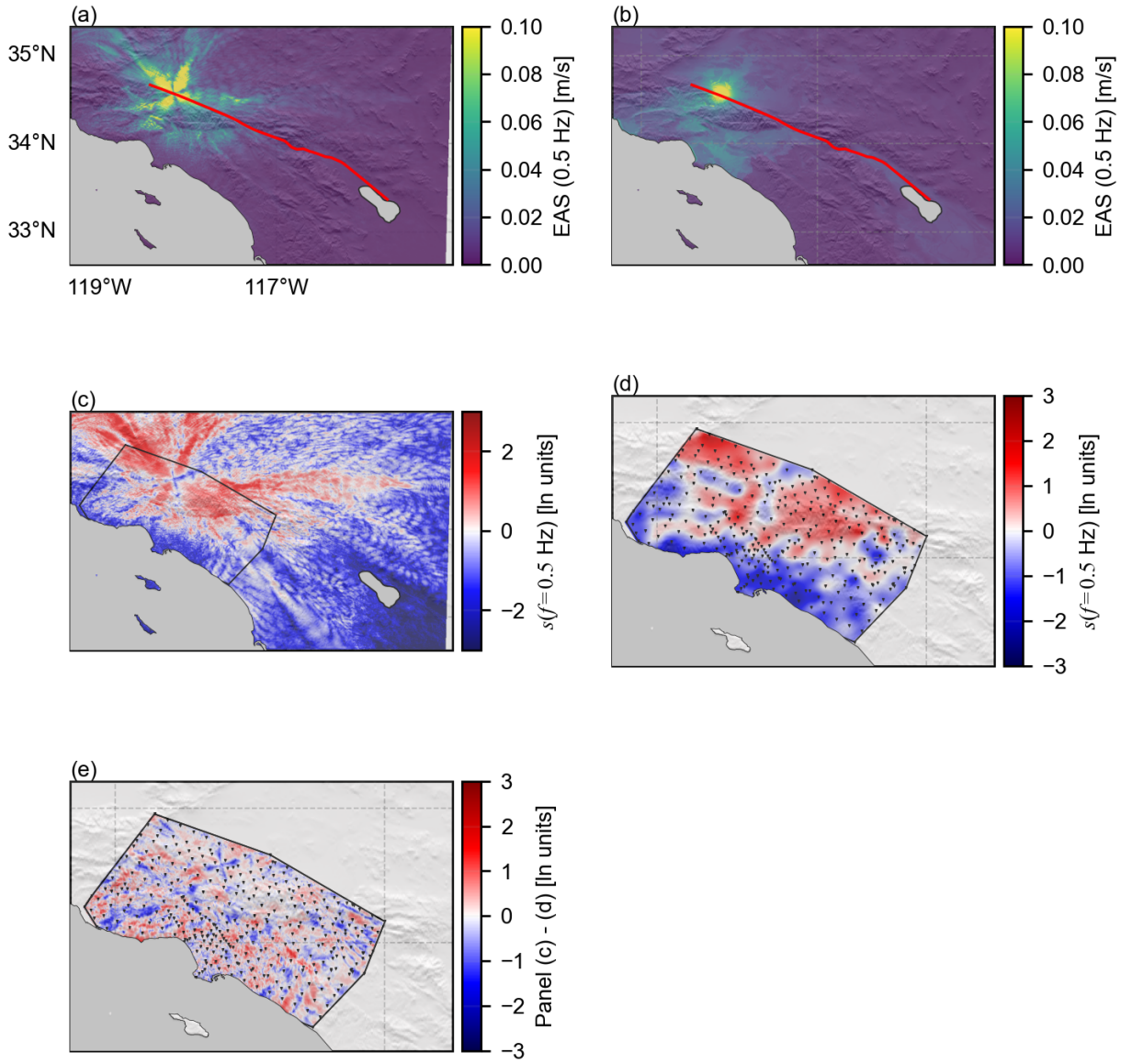


Figure 4: (a) Effective amplitude spectra (EAS) at 0.5 Hz computed from simulated seismogram data located in Southern California with 500 m spatial resolution. The results are shown for a point source with $M_W = 5.0$. The surface trace of the SSAF is indicated in red. (b) EAS predicted from the BA18 GMM using the site parameters computed using the same velocity model that we use in our simulations. (c) Centered residual between the simulations and GMM, measured in natural log units. The polygon indicates the boundary of the convex hull of the CyberShake 22.12 station locations. (d) Interpolation of the residuals using only the data at the CyberShake 22.12 station locations marked by black triangles. The interpolation uses the minimum curvature with tension algorithm from GMT. (e) Difference between panels (c) and (d), i.e., the high-resolution residuals minus the interpolated residuals.

Alt-text: Five-panel map showing 0.5 Hz ground motion results for a point source. Panels show (a) simulated EAS, (b) GMM-predicted EAS, (c) simulation-GMM residuals, (d) interpolated residuals from CyberShake stations marked by triangles, and (e) difference between high-resolution and interpolated residuals.

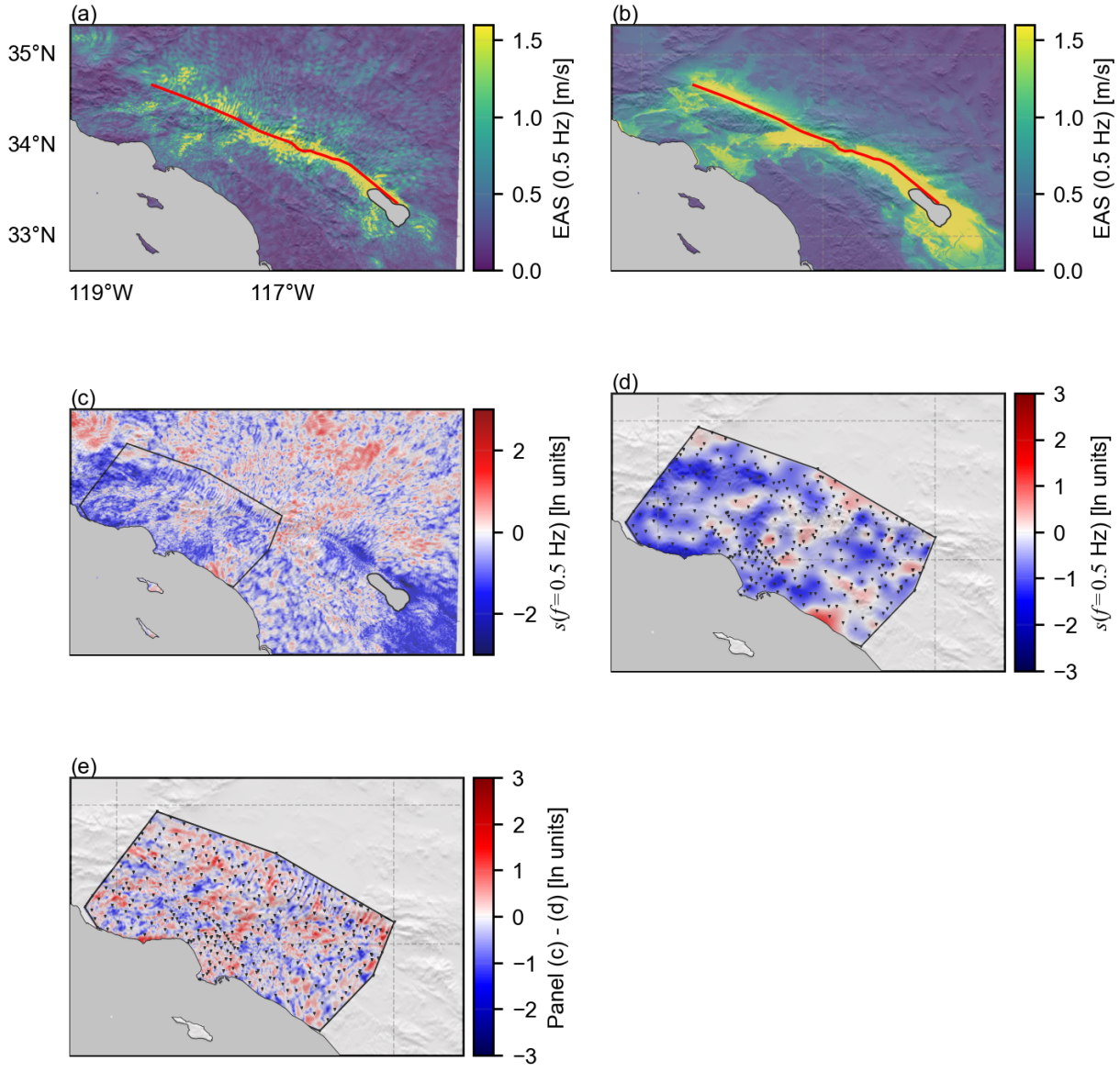


Figure 5: Same as Figure 4, but for the SSAF kinematic rupture model shown in Figure 3.

Alt-text: Five-panel map showing 0.5 Hz ground motion results for a kinematic rupture model. Layout identical to point source figure with panels for simulated EAS, GMM prediction, residuals, interpolated residuals, and interpolation errors across Southern California.

and spatial patterns are comparable to the point-source cases, suggesting that sparse spatial sampling is the dominant source of error rather than the source complexity itself.

Frequency-dependence and the comparison between point source and kinematic rupture model results

When constructing seismic hazard maps, it is important to consider the range of frequencies that affect hazard differently. Our simulated range is from 0.1 to 1.0 Hz, and we select four frequencies (0.1, 0.25, 0.5, and 1.0 Hz) in this range to show, such as Figure 6 which shows the interpolation error averaged across the 100 point sources. At 0.1 Hz, the interpolation errors tend to be largest near transitions between basins and mountainous topography, or where there is a large spatial gradient for the V_{S30} and $Z_{1.0}$ site parameters. The interpolation performs poorer in these cases as some discontinuities in the basin depth models and the velocity model cause sharper spatial changes. However, these maps are based on the residuals from the BA18 GMM, so these are likely due to effects that are not completely captured by the GMM.

At higher frequencies, such as at 1.0 Hz, the interpolation error seemingly becomes more spatially random. Many fewer locations show error close to zero, except for in very close proximity to the CyberShake defined locations where the error is defined to be zero.

Interestingly, the kinematic rupture results are somewhat similar to the point source results (Figure 7). The differences between point source and kinematic rupture models become larger at higher frequencies, though they remain relatively small at less than 0.5 natural log units (Figure 8). At lower frequencies, the differences between the point source and kinematic rupture models are nonzero only in the near-fault region, likely due to the stochastic variability involved in the slip generation and directivity effects for the kinematic rupture models.

To examine the difference between the hazard corrections for point source and kinematic rupture models, we also look at the variations as a function of frequency (Figure 9). For each frequency between 0.1 and 1.0 Hz, we took the mean of the absolute values of the interpolation error, averaged across all the point sources and rupture models. These errors steadily increase with frequency for both types of sources, from about 0.1 natural log units at 0.1 Hz to 0.2 natural log units at 1.0 Hz. The errors are slightly larger for the kinematic rupture model at all frequencies, though the difference becomes larger starting at approximately 0.5 Hz.

Fillmore Basin site example

To demonstrate an example of a site with ground motion effects that are challenging to model, we select a site in the Fillmore Basin (see Figure 2b for the location in the area of interest). This area, part of the Santa Clara River Valley which connects to the Ventura Basin, shows deep basin structure according to velocities in CVM-S4.M26.01

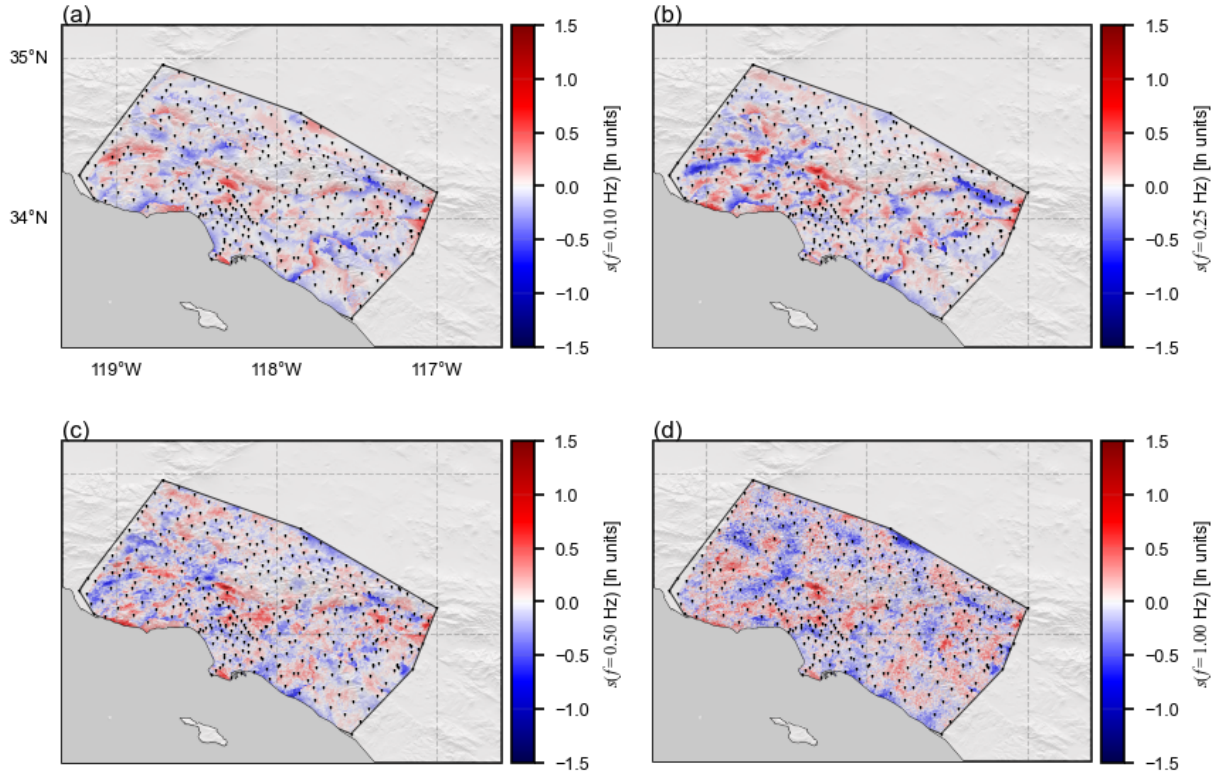


Figure 6: Differences between the high-resolution and the interpolated residual maps for the frequencies 0.1 Hz (a), 0.2 Hz (b), 0.5 Hz (c), and 1.0 Hz (d). The maps show the averaged results across all the 100 point source locations on the SSAF.

Alt-text: Four-panel map showing interpolation errors averaged across 100 point sources at frequencies 0.1, 0.2, 0.5, and 1.0 Hz. Color scale shows natural log unit differences.

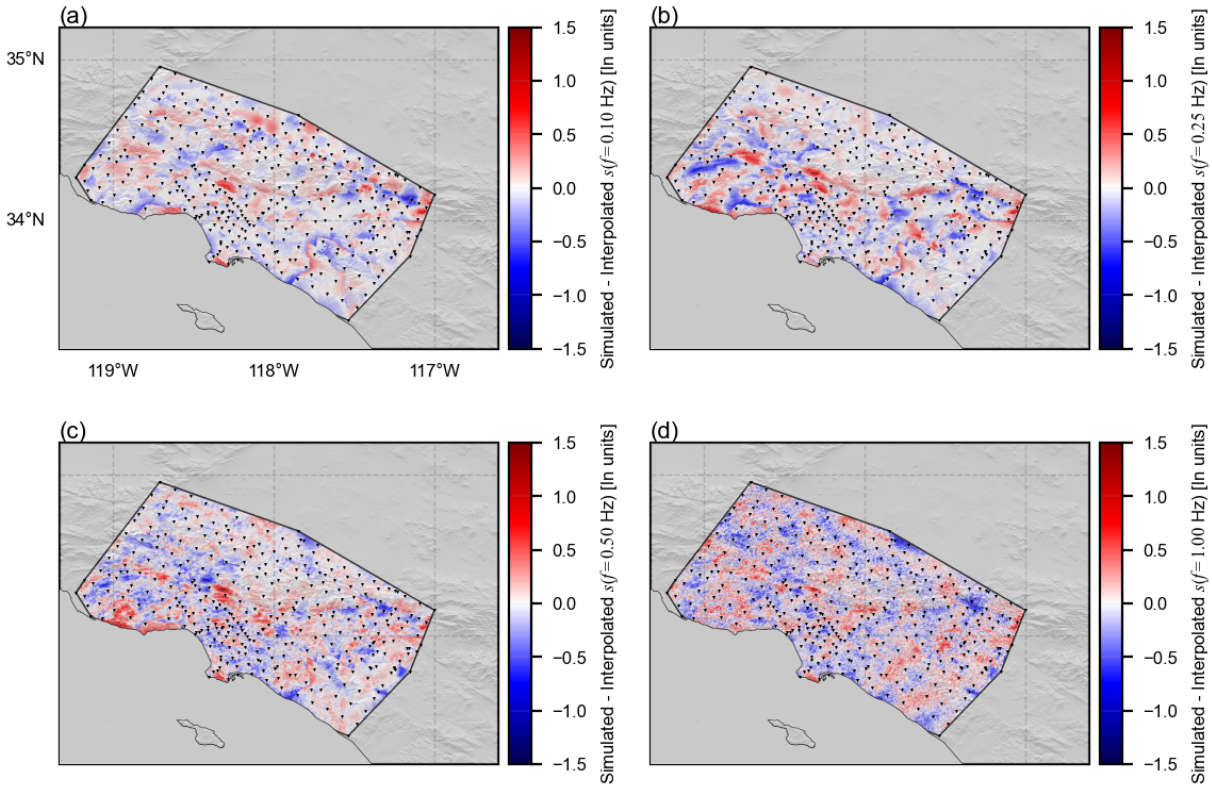


Figure 7: Differences between the high-resolution and the interpolated residual maps for the 100 generated kinematic rupture models.

Alt-text: Four-panel map showing interpolation errors averaged across 100 kinematic rupture models at frequencies 0.1, 0.2, 0.5, and 1.0 Hz. Spatial patterns similar to point source results with increasing errors at higher frequencies.

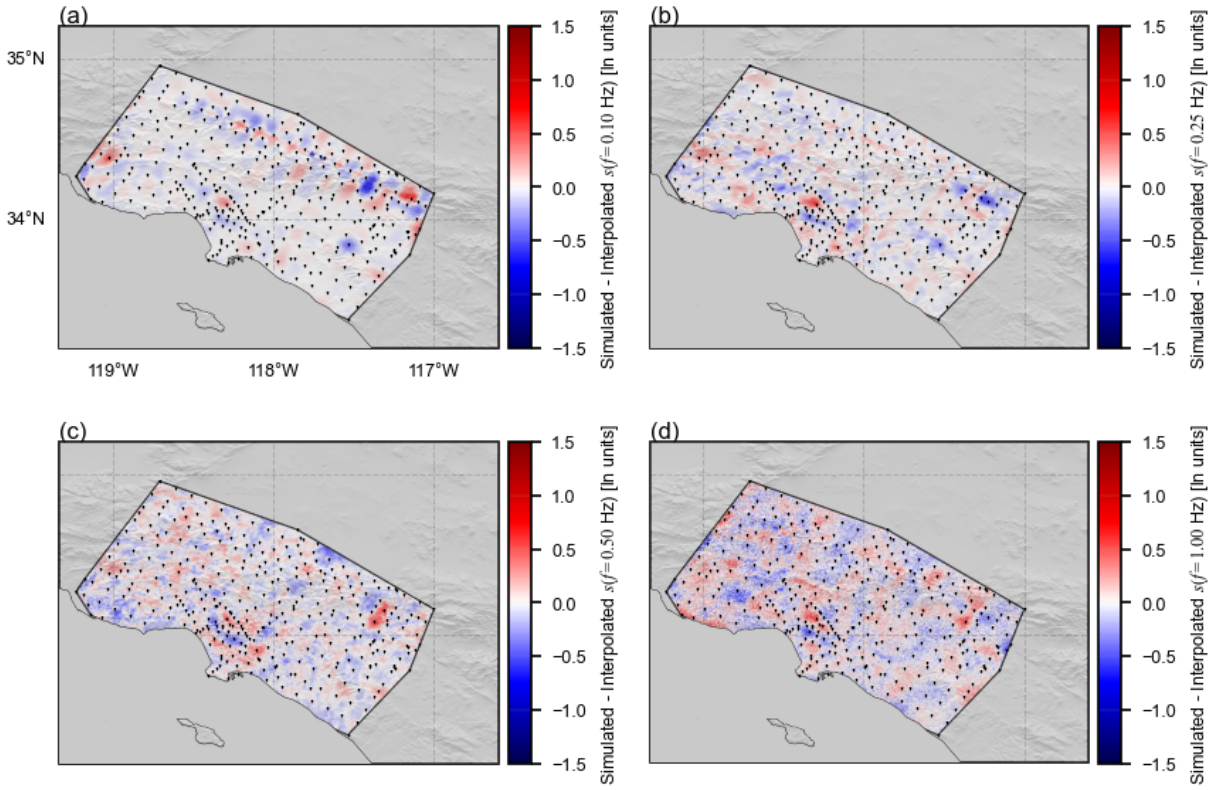


Figure 8: Difference between the results plotted in Figures 6 and 7 (point source interpolation errors - kinematic rupture model interpolation errors).

Alt-text: Four-panel map showing difference between point source and kinematic rupture interpolation errors at four frequencies.

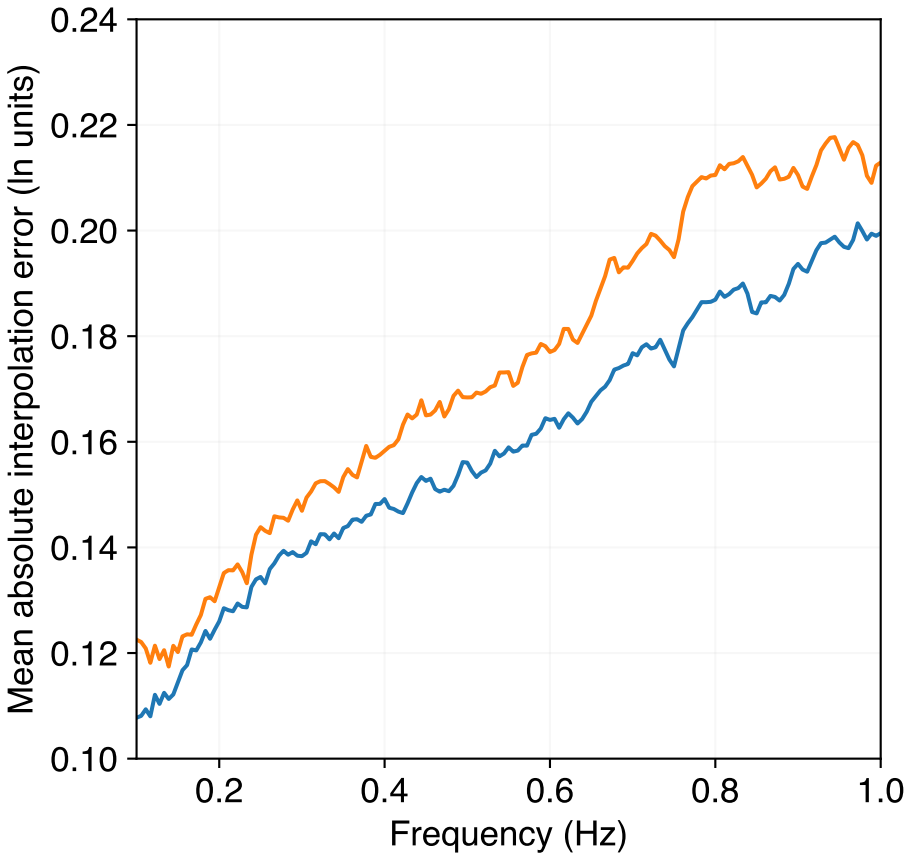


Figure 9: Mean absolute value of interpolation error as a function of frequency for both the point sources (blue) and the kinematic rupture models (orange). At each frequency, the results show the mean absolute difference between the interpolated site residuals and the high-resolution residuals for all receivers and sources. The error trends are similar between the point source and kinematic rupture models, with slightly larger errors for the kinematic rupture models for all frequencies.

Alt-text: Line plot showing mean absolute interpolation error versus frequency from 0.1 to 1.0 Hz. Blue line for point sources and orange line for kinematic rupture models both increase from about 0.1 to 0.2 natural log units, with kinematic rupture models showing slightly larger errors.

with $Z_{1.0}$ reaching up to 1.5 km in the deepest part of the basin. This is also one of the most hazardous locations in the CyberShake 22.12 hazard map, with the 2 s spectral acceleration 2% probability of exceedance in 50 yr at approximately 1 g. Nearby faults in the Western Transverse Ranges Area also contribute to the hazard potential in this area. Some CyberShake stations located in the Fillmore Basin constrain the hazard in this area; however, we observe some errors around 1 natural log unit suggesting that the number of sites is still not sufficient to accurately model the seismic hazard in these types of geologic features.

The simulated EAS are shown in Figure 10a as a function of frequency for all 100 kinematic rupture models. There is a large amount of variation in the EAS depending on the hypocenter location, likely due to the slip variability and directivity effects. However, this station is not directly in line with the strike of the fault where forward and backward directivity would be expected to be strongest. The mean EAS across all rupture models is shown by the thicker red line. The BA18 GMM prediction is shown in blue, and generally underpredicts the mean simulated EAS except at relatively higher frequencies above 0.8 Hz where the simulated EAS decay more rapidly. The wide spread in EAS values at all frequencies demonstrates the significant sensitivity to rupture characteristics, with variations spanning more than an order of magnitude at some frequencies.

The centered residuals, $s(f)$, are shown in Figure 10b for both the high-resolution simulation results (blue lines) and the interpolated values (red line). The interpolated residual consistently under-predicts the simulated residuals at all frequencies when using the minimum curvature interpolation method. The difference between the mean high-resolution centered residual and the interpolated value reaches a maximum of approximately 1.2 natural log units around 0.3 Hz. This indicates a factor of approximately 3.3 in ground motion amplification relative to the site term that would be estimated using the interpolation approach used in CyberShake. This substantial discrepancy highlights the limitations of sparse station spacing in capturing highly local basin amplification effects, particularly in complex geological settings like the Fillmore Basin (e.g., Keller et al., 1982).

Computational cost and storage requirements

For our computational results, the most expensive part of this analysis is performing the upfront waveform modeling and Green's function calculations, which requires 3,920 node-hrs per point source. In comparison, the cost to produce horizontal-component Green's functions with the ROM from 0.1 to 1.0 Hz for a single point source is 0.022 CPUh, and 0.12 CPUh for the kinematic rupture model composed of 110,842 subfaults. Using single precision for both the real and imaginary parts, the digital storage required to store all Green's functions is 125 GB. This is also the same size required to store the complete $\boldsymbol{U}_j$ matrices which contain the left singular vectors needed to produce the ROM Green's functions.

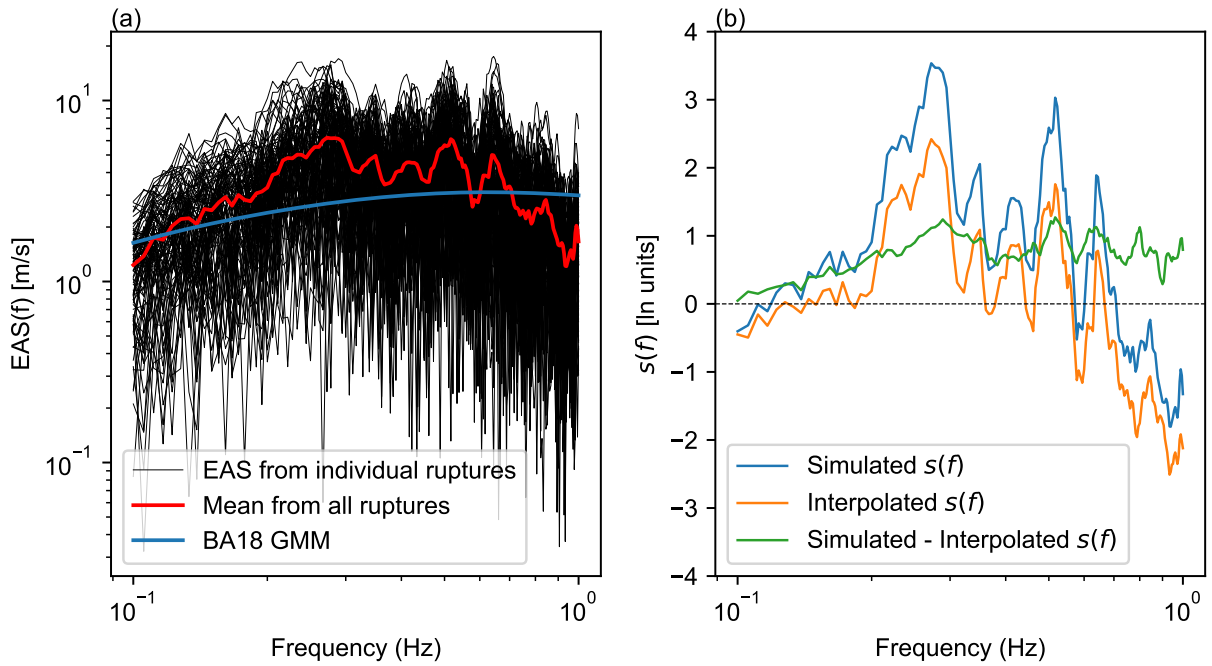


Figure 10: EAS for one location in the Fillmore Basin indicated by the star in Figure 2b. The simulated minus interpolated $s(f)$ is just over one natural log unit for some frequencies.

Alt-text: Plot of effective amplitude spectra versus frequency for Fillmore Basin site. Shows simulated and interpolated curves with difference exceeding one natural log unit at some frequencies, demonstrating interpolation underestimates ground motion at this location.

Discussion

Implications for CyberShake hazard maps

The results that we present here suggest ways that ROMs can inform the expected vibrations in cases where more spatial resolution is warranted. For example, the high-resolution expected vibrations could be used at different points along structures that may be vulnerable to earthquakes, such as bridges or dams, or for distributed infrastructure such as highways, aqueducts, and pipelines, for which hazard can be difficult to capture with traditional PSHA. Recent attention to supershear earthquakes (Goldberg et al., 2025; Wei et al., 2025) also raises questions regarding the possibility of supershear earthquakes along these sections of the SSAF in California. Similarities between the SSAF system and the East Anatolian Fault Zone that hosted the 2023 Turkey-Syria earthquakes which experienced supershear rupture (e.g., Jia et al., 2023; Melgar et al., 2023), also motivate hazard studies that consider these effects. We expect the method developed here is well suited to examine these possibilities with varying rupture speeds or directions of supershear propagation, and to examine detailed wavefields from potential ruptures.

The velocity model, attenuation model, and V_{S30} model are all assumptions that affect the ground motion and hazard result when using physics-based simulations. Finer scale, more local structures in these models may be imaged using dense seismic networks or DAS technology. As a result, using reciprocity and the interpolation approach using minimum curvature with adjustable tension may not be suitable when the hazard correction terms are not smoothly varying between neighboring sites. As the interpolation quality is also poorer for higher frequencies, which are also more computationally expensive to simulate deterministically, this motivates the use of ROMs for forward simulations that would perform efficiently with many sites. The poorer interpolation quality at higher frequencies stems from the shorter wavelengths involved, which cause ground motions to vary more rapidly in space due to small-scale heterogeneities in the velocity structure and basin geometries that cannot be adequately resolved by the sparse station spacing used in CyberShake studies. As a result, hazard maps derived from interpolated site terms may mischaracterize the relative hazard between nearby locations, particularly across basin edges, which has direct implications for infrastructure prioritization. We also find that our results show larger interpolation errors at sharp basin edges and with sharp changes in topography, so calculations that do use reciprocity should consider strategically placing sites to adequately represent these features. One strategy could place more sites at basin edges or where the spatial gradient of site parameter maps is large.

Computational cost comparison between ROM and reciprocity-based approaches

In CyberShake 22.12, the calculations required 140 node-hours on the Summit supercomputer for the SGTs per site, and 250 node-hours for the seismogram synthesis (Graves et al., 2011; Jordan and Callaghan, 2018; Callaghan et al.,

2024). These calculations use the UCERF2 faults and a maximum rupture distance of 200 km which results in more faults in Southern California than just the SSAF (Figure S2). As the CyberShake 22.12 calculations use reciprocity, increasing to 500-meter resolution throughout our domain that requires 480,000 sites would significantly increase computational costs. Assuming the costs would scale up linearly to 500-m resolution, the CyberShake SGT method would require $(140)(480,000) = 67,200,000$ node-hours. In this work, we find that about 7,000 node-hours total are required on the Frontera supercomputer to perform the FOM simulations to obtain the SGTS for a single strike-slip orientation for the SSAF. This number would have been doubled to 14,000 node-hours if we had included dip-slip motion which is included in CyberShake SGT calculations.

Constructing a complete hazard map for the region would require considering all sources in the earthquake rupture forecast. Based on the amount of SSAF fault area out of the area of all faults in the UCERF2 faults within the maximum distance, we calculate that our area of the SSAF surface accounts for 7% of the UCERF2 fault surface area. Using the iPOD approach for all UCERF2 faults, and assuming a uniform spatial sampling density across all faults, we expect this would require $(14,000)/(0.07) = 200,000$ node-hours. This estimate assumes that other faults in the rupture forecast could be adequately sampled using the same number of sources per fault area as the SSAF. This cost estimate is significantly lower than the reciprocity-based estimate, with a ROM speed-up factor of $(67,200,000)/(200,000) = 336$, suggesting that high spatial resolution in hazard maps can be achieved without incurring extreme computational costs. Previous studies provided rules-of-thumb for estimating when it is more efficient to use reciprocal or forward simulations of wave propagation, depending on the number of sites and the number of seismic sources (e.g., Graves and Wald, 2001). This work confirms that using some forward simulations to first construct accurate ROMs, and then using them for hazard calculations is more efficient when the number of sites becomes large. This work does not consider constructing ROMs using iPOD from reciprocal simulations due to local site variations at the surface that are expected to cause poor interpolation. We also note that hazard maps derived from ROMs of seismic wave propagation will not be exact solutions. The relative error between the ROM and FOM in the EAS predictions for single point sources is about 25 percent for the entire domain, but it is not possible to quantify error for all of the rupture scenarios used to construct the hazard map. We also do not suggest that our maps provide true corrections to hazard maps, as corrections would require including the bias and event terms when using the current CyberShake method with GMMs.

We also expect that differences in the hardware between Frontera and Summit used for the SGT calculations would impact the direct comparison of computational costs, though these hardware factors would not be large enough to change our conclusions. Similarly, the CyberShake SGT calculation uses the AWP-ODC software which is based on the staggered finite difference method (Cui et al., 2010) compared to SeisSol which uses ADER-DG.

Future developments

There are a few other ways in which our approach could be further developed. One way is that we chose to only use a single focal mechanism to represent the average strike-slip motion on the SSAF. However, in the kinematic rupture model generators, which is also something observed in dynamic rupture simulations, there can be slip motion that deviates from the strike-slip motion, either due to geometric complexities, roughness, or other dynamic effects (e.g., Bizzarri and Cocco, 2005; Guatteri and Spudich, 1998). Another limitation is that we only considered one branch of the SSAF, and our 2D approximation using only the distance from a reference point and the hypocentral depth would break down for faults that exhibit many branches or for ones that highly deviate from a straight line approximation. We expect that our method would apply for non strike-slip earthquakes as well, such as the less frequent earthquakes on the Puente Hills thrust fault. The ROM approach could also be used directly to improve interpolation in the existing CyberShake reciprocity calculation pipeline. For example, high-resolution ROM site amplification maps could be produced for each source, and then used to interpolate CyberShake ground motions between sites (rather than the existing spline interpolation). We also do not consider the possibility of different interpolators, such as the conditional multivariate normal distribution used by ShakeMap (Worden et al., 2018) to estimate ground motion where observations are not available.

Conclusion

Our conclusions are as follows: (1) The CyberShake interpolation of site terms using minimum curvature with tension will over-predict and under-predict probabilistic seismic hazard for earthquake ruptures in between CyberShake site locations. The CyberShake 22.12 study shows residuals up to a factor of approximately three compared to our high-resolution wavefield modeling, with the residuals being noticeably larger near basin edges, and with mean absolute errors tending to increase with increasing frequency from 0.1 to 1.0 Hz. (2) Using ROMs to obtain Fourier amplitude spectra from a set of kinematic rupture models show minor differences compared to an average of point source simulations for the Fourier amplitude spectra, suggesting that hazard maps using Fourier amplitudes could be obtained rapidly provided that enough sampling of the fault surfaces is performed. However, ground motion effects in the near-field would require considering the effects from the kinematic rupture models, especially if there is a preferred rupture direction affecting directivity. Other intensity metrics, such as peak ground velocity, peak ground acceleration, and pseudospectral accelerations could also be computed using reduced-order modeling of the phase and amplitude information from our Green's functions. (3) Future applications in CyberShake exploring higher frequency wavefields should incorporate forward simulations (i.e., not using reciprocity) which may be performed quickly for obtaining physics-informed wavefields in the frequency domain.

Data and Resources

The Supplemental Material to this article includes additional figures that show the site parameter maps, the faults used in hazard calculations, as well as a table of the inputs to the rupture generator. The SeisSol input files and FAS data are available in a private Zenodo repository at this link: https://zenodo.org/records/18249723?preview=1&token=eyJhbGciOiJIUzUxMiJ9.eyJpZCI6IjhjZGIwODZlLTU1MmYtNGUwNC1iMmIwLTE5OTc1YjhhM2YxZCIsImRhdGEiOnt9LCJy_p1V44zAIGnjRoDDHT_2wVmBCXd2P29eBSYvE_H9YqjgTDFBZZgjssCmVtg8mpQWa4iKeN2HBdSmfdISUJ5WHA The authors are not able to view who has accessed this link. The repository also includes a Jupyter notebook that demonstrates how to access the data. The repository will be made public upon acceptance of this paper. The SeisSol code repository can be found at https://github.com/SeisSol/SeisSol and the CyberShake code can be found at https://github.com/SCECcode/cybershake-core.

Declaration of Competing Interests

The authors acknowledge there are no conflicts of interest recorded.

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References

Azar, S. and M. Dabaghi (2021). Simulation-Based Seismic Hazard Assessment Using Monte-Carlo Earthquake Catalogs: Application to CyberShake. *Bulletin of the Seismological Society of America* **111**(3), 1481–1493.

Bayless, J. and N. A. Abrahamson (2019). Summary of the BA18 ground-motion model for Fourier amplitude spectra for crustal earthquakes in California. *Bulletin of the Seismological Society of America* **109**(5), 2088–2105.

Beyer, J., M. L. Cooke, and S. T. Marshall (2018). Sensitivity of deformation to activity along the Mill Creek and Mission Creek strands of the southern San Andreas fault. *Geosphere* **14**(6), 2296–2310.

Bizzarri, A. and M. Cocco (2005). 3D dynamic simulations of spontaneous rupture propagation governed by different constitutive laws with rake rotation allowed. *Annals of Geophysics* **48**(2), 279–299.

Breuer, A. and A. Heinecke (2022). Next-generation local time stepping for the ADER-DG finite element method. In *2022 IEEE international parallel and distributed processing symposium (IPDPS)*, pp. 402–413. IEEE.

Bui-Thanh, T., M. Damodaran, and K. Willcox (2003). Proper orthogonal decomposition extensions for parametric applications in compressible aerodynamics. In *21st AIAA applied aerodynamics conference*, pp. 4213.

Callaghan, S., P. J. Maechling, F. Silva, M.-H. Su, K. R. Milner, R. W. Graves, K. B. Olsen, Y. Cui, K. Vahi, A. Kottke, et al. (2024). Using open-science workflow tools to produce scec cybershake physics-based probabilistic seismic hazard models. *Frontiers in High Performance Computing* **2**, 1360720.

Cui, Y., K. B. Olsen, T. H. Jordan, K. Lee, J. Zhou, P. Small, D. Roten, G. Ely, D. K. Panda, A. Chourasia, et al. (2010). Scalable earthquake simulation on petascale supercomputers. In *SC'10: Proceedings of the 2010 ACM/IEEE International Conference for High Performance Computing, Networking, Storage and Analysis*, pp. 1–20. IEEE.

Cui, Y., E. Poyraz, K. B. Olsen, J. Zhou, K. Withers, S. Callaghan, J. Larkin, C. Guest, D. Choi, A. Chourasia, et al. (2013). Physics-based seismic hazard analysis on petascale heterogeneous supercomputers. In *Proceedings of the international conference on high performance computing, networking, storage and analysis*, pp. 1–12.

Druault, P., P. Guibert, and F. Alizon (2005). Use of proper orthogonal decomposition for time interpolation from PIV data: Application to the cycle-to-cycle variation analysis of in-cylinder engine flows. *Experiments in fluids* **39**(6), 1009–1023.

Dumbser, M. and M. Käser (2006). An arbitrary high-order discontinuous Galerkin method for elastic waves on unstructured meshes—II. The three-dimensional isotropic case. *Geophysical Journal International* **167**(1), 319–336.

Ely, G. P., T. Jordan, P. Small, and P. J. Maechling (2010). A Vs30-derived near-surface seismic velocity model. In *Abstract S51A-1907, fall meeting*. AGU San Francisco, CA.

Field, E. H., R. J. Arrowsmith, G. P. Biasi, P. Bird, T. E. Dawson, K. R. Felzer, D. D. Jackson, K. M. Johnson, T. H. Jordan, C. Madden, et al. (2014). Uniform California earthquake rupture forecast, version 3 (UCERF3)—The time-independent model. *Bulletin of the Seismological Society of America* **104**(3), 1122–1180.

Field, E. H., G. P. Biasi, P. Bird, T. E. Dawson, K. R. Felzer, D. D. Jackson, K. M. Johnson, T. H. Jordan, C. Madden, A. J. Michael, et al. (2015). Long-term time-dependent probabilities for the third uniform california earthquake rupture forecast (ucerf3). *Bulletin of the Seismological Society of America* **105**(2A), 511–543.

Folk, M., G. Heber, Q. Koziol, E. Pourmal, and D. Robinson (2011). An overview of the HDF5 technology suite and its applications. In *Proceedings of the EDBT/ICDT 2011 workshop on array databases*, pp. 36–47.

Gerstenberger, M. C., S. Bora, B. A. Bradley, C. DiCaprio, A. Kaiser, E. F. Manea, A. Nicol, C. Rollins, M. W. Stirling, K. K. Thingbaijam, et al. (2024). The 2022 Aotearoa New Zealand national seismic hazard model: Process, overview, and results. *Bulletin of the Seismological Society of America* **114**(1), 7–36.

Goldberg, D. E., W. L. Yeck, C. E. Hanagan, J. Atterholt, H. Kehoe, N. G. Reitman, W. D. Barnhart, D. R. Shelly, A. E. Hatem, D. Wald, et al. (2025). Ultralong, supershear rupture of the 2025 Mw 7.7 Mandalay earthquake reveals unaccounted risk. *Science* **390**(6772), 458–462.

Goulet, C., A. Kottke, D. Boore, Y. Bozorgnia, J. Hollenback, T. Kishida, A. Der Kiureghian, O. Ktenidou, N. Kuehn, E. Rathje, et al. (2018). Effective amplitude spectrum (EAS) as a metric for ground motion modeling using Fourier amplitudes. In *2018 Seismology of the Americas Meeting*.

Graves, R., T. H. Jordan, S. Callaghan, E. Deelman, E. Field, G. Juve, C. Kesselman, P. Maechling, G. Mehta, K. Milner, et al. (2011). CyberShake: A physics-based seismic hazard model for southern California. *Pure and Applied Geophysics* **168**(3), 367–381.

Graves, R. and A. Pitarka (2016). Kinematic Ground-Motion Simulations on Rough Faults Including Effects of 3D Stochastic Velocity Perturbations. *Bulletin of the Seismological Society of America* **106**(5), 2136–2153.

Graves, R. W., B. T. Aagaard, and K. W. Hudnut (2011). The ShakeOut earthquake source and ground motion simulations. *Earthquake Spectra* **27**(2), 273–291.

Graves, R. W. and D. J. Wald (2001). Resolution analysis of finite fault source inversion using one-and three-dimensional Green's functions: 1. Strong motions. *Journal of Geophysical Research: Solid Earth* **106**(B5), 8745–8766.

Guatteri, M. and P. Spudich (1998). Coseismic temporal changes of slip direction: the effect of absolute stress on dynamic rupture. *Bulletin of the Seismological Society of America* **88**(3), 777–789.

Halton, J. H. (1960). On the efficiency of certain quasi-random sequences of points in evaluating multi-dimensional integrals. *Numerische Mathematik* **2**(1), 84–90.

Harris, C. R., K. J. Millman, S. J. Van Der Walt, R. Gommers, P. Virtanen, D. Cournapeau, E. Wieser, J. Taylor, S. Berg, N. J. Smith, et al. (2020). Array programming with NumPy. *Nature* **585**(7825), 357–362.

Heinecke, A., A. Breuer, S. Rettenberger, M. Bader, A.-A. Gabriel, C. Pelties, A. Bode, W. Barth, X.-K. Liao, K. Vaidyanathan, et al. (2014). Petascale high order dynamic rupture earthquake simulations on heterogeneous

supercomputers. In *SC'14: Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis*, pp. 3–14. IEEE.

Hudnut, K. W., L. Seeber, and J. Pacheco (1989). Cross-fault triggering in the November 1987 Superstition Hills earthquake sequence, southern California. *Geophysical Research Letters* **16**(2), 199–202.

Jia, Z., Z. Jin, M. Marchandon, T. Ulrich, A.-A. Gabriel, W. Fan, P. Shearer, X. Zou, J. Rekoske, F. Bulut, A. Garagon, and Y. Fialko (2023). The complex dynamics of the 2023 Kahramanmaraş, Turkey, Mw 7.8-7.7 earthquake doublet. *Science* **381**(6661), 985–990.

Jordan, T. H. and S. Callaghan (2018). Cybershake models of seismic hazards in southern and central california. In *Proceedings of the US national conference on earthquake engineering*.

Käser, M. and M. Dumbser (2006). An arbitrary high-order discontinuous Galerkin method for elastic waves on unstructured meshes—I. The two-dimensional isotropic case with external source terms. *Geophysical Journal International* **166**(2), 855–877.

Käser, M., M. Dumbser, J. De La Puente, and H. Igel (2007). An arbitrary high-order discontinuous Galerkin method for elastic waves on unstructured meshes—III. Viscoelastic attenuation. *Geophysical Journal International* **168**(1), 224–242.

Keller, E. A., D. Johnson, M. Clark, and T. Rockwell (1982). Tectonic geomorphology and earthquake hazard, north flank, central ventura basin, california. Technical report, US Geological Survey,.

Kikuchi, M. and H. Kanamori (1991). Inversion of Complex Body Waves—III. *Bulletin of the Seismological Society of America* **81**(6), 2335–2350.

Kishida, T., O.-J. Ktenidou, R. B. Darragh, and S. Walter (2016). Semi-automated procedure for windowing time series and computing Fourier amplitude spectra (FAS) for the NGA-West2 database. Technical report.

Melgar, D., T. Taymaz, A. Ganas, B. Crowell, T. Öcalan, M. Kahraman, V. Tsironi, S. Yolsal-Çevikbilen, S. Valkaniotis, T. S. Irmak, T. Eken, C. Erman, B. Özkan, A. H. Dogan, and C. Altuntaş (2023). Sub- and Super-Shear Ruptures during the 2023 Mw 7.8 and Mw 7.6 Earthquake Doublet in SE Türkiye. *Seismica* **2**(3).

Milner, K. R., B. E. Shaw, C. A. Goulet, K. B. Richards-Dinger, S. Callaghan, T. H. Jordan, J. H. Dieterich, and E. H. Field (2021). Toward physics-based nonergodic PSHA: A prototype fully deterministic seismic hazard model for southern California. *Bulletin of the Seismological Society of America* **111**(2), 898–915.

Olsen, K. B., S. M. Day, and C. R. Bradley (2003). Estimation of Q for Long-Period (>2 sec) Waves in the Los Angeles Basin. *Bulletin of the Seismological Society of America* **93**(2), 627–638.

Petersen, M. D., A. M. Shumway, P. M. Powers, E. H. Field, M. P. Moschetti, K. S. Jaiswal, K. R. Milner, S. Rezaeian, A. D. Frankel, A. L. Llenos, et al. (2024). The 2023 US 50-state national seismic hazard model: Overview and implications. *Earthquake Spectra* **40**(1), 5–88.

Rekoske, J. M., A.-A. Gabriel, and D. A. May (2023). Instantaneous physics-based ground motion maps using reduced-order modeling. *Journal of Geophysical Research: Solid Earth* **128**(8), e2023JB026975.

Rekoske, J. M., D. A. May, and A.-A. Gabriel (2025). Reduced-order modelling for complex three-dimensional seismic wave propagation. *Geophysical Journal International* **241**(1), 526–548.

Rettenberger, S., O. Meister, M. Bader, and A.-A. Gabriel (2016). Asagi: A parallel server for adaptive geoinformation. In *Proceedings of the Exascale Applications and Software Conference 2016*, pp. 1–9.

Rodgers, A. J., A. Pitarka, R. Pankajakshan, B. Sjögreen, and N. A. Petersson (2020). Regional-Scale 3D Ground-Motion Simulations of Mw 7 Earthquakes on the Hayward Fault, Northern California Resolving Frequencies 0–10 Hz and Including Site-Response Corrections. *Bulletin of the Seismological Society of America* **110**(6), 2862–2881.

Shaw, B. E., K. R. Milner, E. H. Field, K. Richards-Dinger, J. J. Gilchrist, J. H. Dieterich, and T. H. Jordan (2018). A physics-based earthquake simulator replicates seismic hazard statistics across California. *Science advances* **4**(8), eaau0688.

Shaw, B. E., K. R. Milner, and C. A. Goulet (2025). Deterministic Physics-Based Earthquake Sequence Simulators Match Empirical Ground-Motion Models and Enable Extrapolation to Data-Poor Regimes: Application to Multi-fault Multimechanism Ruptures. *Seismological Research Letters* **96**(4), 2431–2444.

Small, P., D. Gill, P. J. Maechling, R. Taborda, S. Callaghan, T. H. Jordan, K. B. Olsen, G. P. Ely, and C. Goulet (2017). The SCEC unified community velocity model software framework. *Seismological Research Letters* **88**(6), 1539–1552.

Smith, W. H. F. and P. Wessel (1990). Gridding with continuous curvature splines in tension. *Geophysics* **55**(3), 293–305.

Stafford, P. J. (2014). Crossed and nested mixed-effects approaches for enhanced model development and removal of the ergodic assumption in empirical ground-motion models. *Bulletin of the Seismological Society of America* **104**(2), 702–719.

Stuart, W. D. (1986). Forecast model for large and great earthquakes in southern California. *Journal of Geophysical Research: Solid Earth* **91**(B14), 13771–13786.

Thompson, E. M. (2022). An Updated Vs30 Map for California with Geologic and Topographic Constraints.

Thompson, E. M., L. G. Baise, Y. Tanaka, and R. E. Kayen (2012). A taxonomy of site response complexity. *Soil Dynamics and Earthquake Engineering* **41**, 32–43.

Thompson, E. M., D. J. Wald, and C. B. Worden (2014). A VS30 Map for California with Geologic and Topographic Constraints. *Bulletin of the Seismological Society of America* **104**(5), 2313–2321.

Tian, D., L. Uieda, W. J. Leong, Y. Fröhlich, M. Grund, W. Schlitzer, M. Jones, L. Toney, J. Yao, J.-H. Tong, Y. Magen, K. Materna, A. Belem, T. Newton, A. Anant, M. Ziebarth, J. Quinn, and P. Wessel (2025). PyGMT: A Python interface for the Generic Mapping Tools.

Uphoff, C. and M. Bader (2016). Generating high performance matrix kernels for earthquake simulations with viscoelastic attenuation. In *2016 International Conference on High Performance Computing & Simulation (HPCS)*, pp. 908–916. IEEE.

Uphoff, C., S. Rettenberger, M. Bader, E. H. Madden, T. Ulrich, S. Wollherr, and A.-A. Gabriel (2017). Extreme scale multi-physics simulations of the tsunamigenic 2004 sumatra megathrust earthquake. In *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis*, pp. 1–16.

Wei, S., X. Wang, C. Li, H. Zeng, Z. Ma, Q. Shi, H. Chen, Y. Huang, M. Lyu, J. Liao, S. Yang, Y. Bai, P. M. Maung, K. Oo, Y. M. M. Htwe, J. Zhang, L. D. Zilio, X. Shan, and L. Chen (2025). Supershear rupture sustained through a thick fault zone in the 2025 M_w 7.8 Mandalay earthquake. *Science* **390**(6772), 468–475.

Wessel, P., J. F. Luis, L. a. Uieda, R. Scharroo, F. Wobbe, W. H. Smith, and D. Tian (2019). The Generic Mapping Tools Version 6. *Geochemistry, Geophysics, Geosystems* **20**(11), 5556–5564.

Worden, C. B., E. M. Thompson, J. W. Baker, B. A. Bradley, N. Luco, and D. J. Wald (2018). Spatial and spectral interpolation of ground-motion intensity measure observations. *Bulletin of the Seismological Society of America* **108**(2), 866–875.

Zhao, L., P. Chen, and T. H. Jordan (2006). Strain green's tensors, reciprocity, and their applications to seismic source and structure studies. *Bulletin of the Seismological Society of America* **96**, 1753–1763.

List of Figures

1 Conceptual figure to illustrate the motivation to improve the spatial resolution in physics-based seismic hazard assessment. Assuming a crustal velocity model, as shown in the cross-section for a selected vertical slice through the Southern California CVM-S4.M26.01, heterogeneities in the subsurface V_S structure result in many different velocity profiles. Adequate resolution of receivers at the surface is required to capture the wavefield effects and resulting hazard. Lower resolution maps, such as using the triangles as sites, are computationally cheaper but not as accurate as using higher resolution sites shown by the blue circles. 4

2 Maps of the study area in Southern California. (a) shows the UCERF3 fault traces indicated by gray lines, and the portion of the SSAF that we use is shown in red. The blue rectangle indicates the boundary of the area where we compute high resolution ground motion site terms for the SSAF ruptures. The CyberShake Study 22.12 sites in Southern California are indicated by the black triangles. (b) shows a zoomed in view of the CyberShake Study 22.12 sites. The gold star indicates a location in the narrow Fillmore Basin where hazard is challenging to model accurately. 6

3 A single kinematic rupture model for the SSAF generated using v5.5.2 of the Graves and Pitarka (2016) kinematic rupture generator. The hypocenter location is indicated by a gold star. The black contours indicate the initiation time of slip in increments of 5 s. 99 other stochastic kinematic rupture models for the same fault geometry are also generated with varying hypocentral locations. 8

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Appendix

Table 1: Alphabetical list of the acronyms used throughout the paper.

Acronym	Meaning
ADER-DG	arbitrary high-order accurate derivative discontinuous Galerkin
AWP-ODC	Anelastic Wave Propagation - Olsen, Day, Cui
API	Application Programming Interface
ASCII	American Standard Code for Information Interchange
CFM	community fault model
CPUh	central processing unit hours
CVM	community velocity model
DAS	distributed acoustic sensing
EAS	effective amplitude spectrum
FAS	Fourier amplitude spectrum
FOM	full-order model
GMM	ground motion model
GMT	Generic Mapping Tools
GPU	graphics processing unit
HDF	Hierarchical Data Format
HPC	high-performance computing
NetCDF	Network Common Data Form
NRF	NetCDF rupture format
NSHM	National Seismic Hazard Model
PGA	peak ground acceleration
PGV	peak ground velocity
POD	proper orthogonal decomposition
SCEC	Statewide California Earthquake Center
ROM	reduced-order model
SGT	strain Green's tensor
SRF	standard rupture format
SSAF	Southern San Andreas Fault
SVD	singular value decomposition
PSHA	probabilistic seismic hazard analysis
UCERF	Uniform California Earthquake Rupture Forecast